Sensor Stress Analyzer v1.0.0

Simulation Results

mode: FEM Analysis

n: 12

force: 50

heat: 50

mesh: {'n': '12', 'vertices': '[[1.0, 0.0], [0.8660254037844387, 0.49999999999999994], [0.5000000000000001, 0.8660254037844386], [6.123233995736766e-17, 1.0], [-0.4999999999999998, 0.8660254037844387], [-0.8660254037844385, 0.5000000000000003], [-1.0, 1.2246467991473532e-16], [-0.8660254037844388, -0.4999999999999997], [-0.5000000000000004, -0.8660254037844385], [-1.8369701987210297e-16, -1.0], [0.49999999999999933, -0.866025403784439], [0.8660254037844384, -0.5000000000000004]]', 'edges': '[[0, 1], [1, 2], [2, 3], [3, 4], [4, 5], [5, 6], [6, 7], [7, 8], [8, 9], [9, 10], [10, 11], [11, 0]]'}

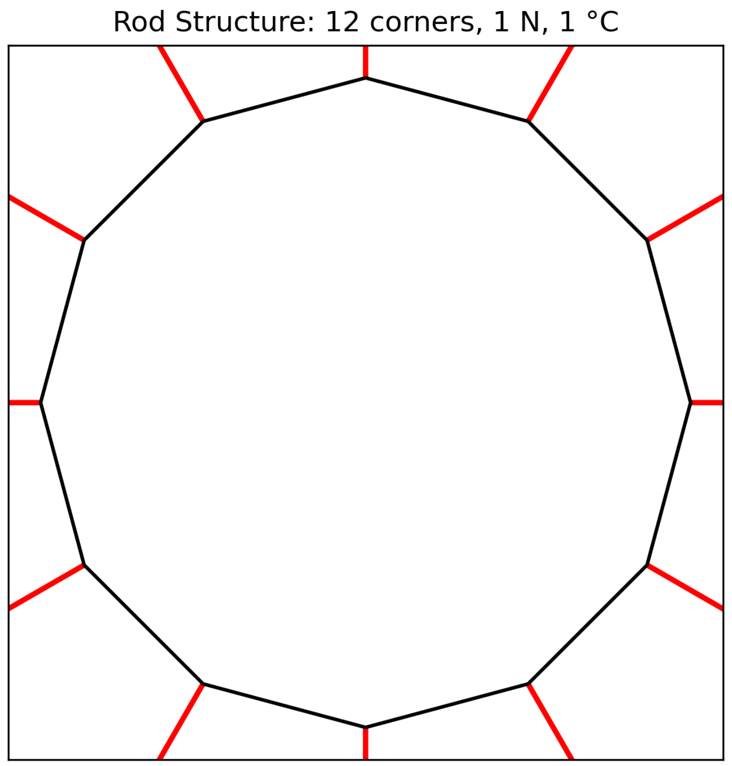
stress_map: [50.0, 52.5, 54.33012701892219, 55.00000000000001, 54.33012701892219, 52.5, 50.0, 47.5, 45.66987298107781, 45.0, 45.6698729810778, 47.5]

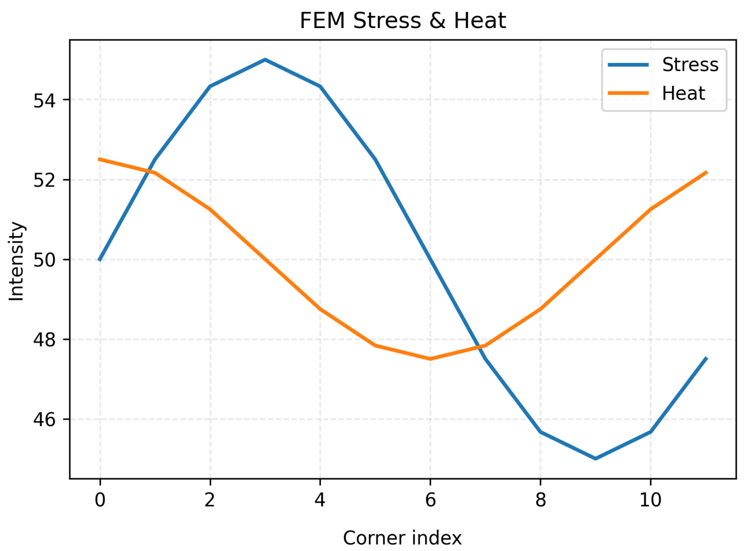
heat_map: [52.5, 52.1650635094611, 51.24999999999999, 50.0, 48.75, 47.8349364905389, 47.5, 47.8349364905389, 48.75, 50.0, 51.24999999999999, 52.1650635094611]

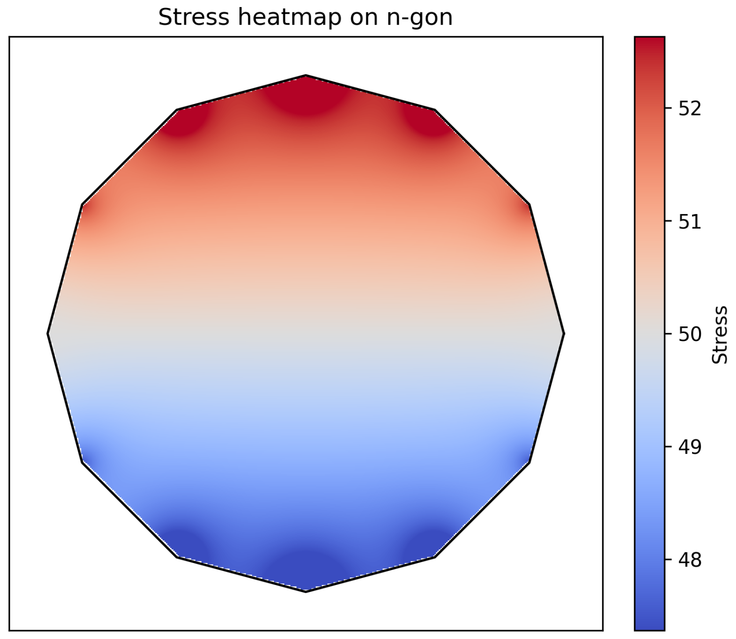
stress_max: 55.00000000000001

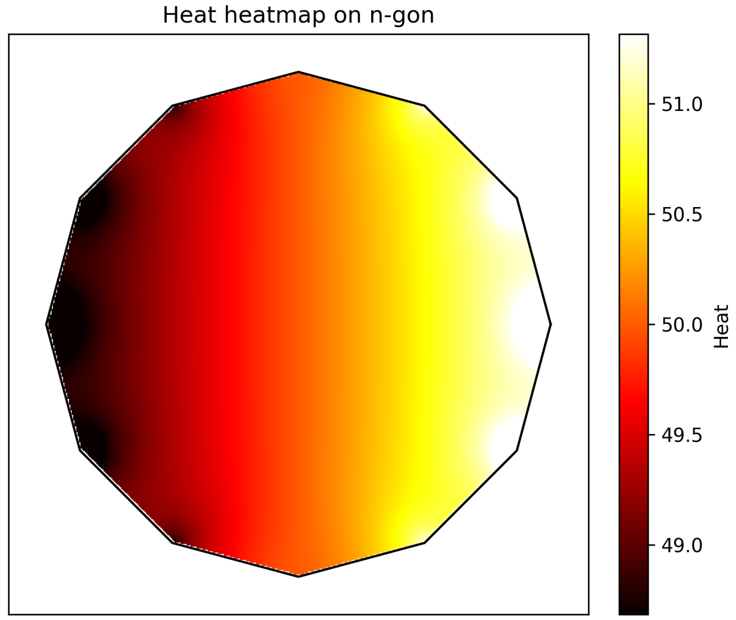
heat_max: 52.5

Simulation Plots


Rod Structure: 12 corners, 1 N, 1 °C


FEM Stress & Heat
Stress
Heat
Intensity
54
52
50
48
46
0
2
4
6
8
10
Corner index





Explainable AI Summary

Sensor Stress Analyzer v1.0.0 - Quantitative Summary

In FEM Analysis mode, the analysis highlights stress and thermal resilience across the sensor housing.

The polygon had 12 corners, subjected to 50 N force and 50 °C heat.

The maximum stress reached 55.00 units.

The maximum heat intensity was 52.50 units.

These values highlight the most critical regions for structural integrity and thermal resilience.